

2.3.17

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Question

Find the properties of the triangle with vertices $(4, 0)$, $(0, 3)$ and $(4, 0)$.

Theoretical Solution

Given the points,

Name	Point
A	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$
B	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Table: Variables used.

Theoretical Solution

We can calculate the magnitude of sides of the triangle using norms:

$$(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (1)$$

$$(\mathbf{B} - \mathbf{C}) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (2)$$

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (3)$$

Checking for orthogonality

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 3 \end{pmatrix} = 16 - 9 = 7 \neq 0 \quad (4)$$

$$(\mathbf{B} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 & 3 \end{pmatrix} \cdot \begin{pmatrix} -8 \\ 0 \end{pmatrix} = 32 + 0 = 32 \neq 0 \quad (5)$$

$$(\mathbf{A} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -8 & 0 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -3 \end{pmatrix} = 32 - 0 = 32 \neq 0 \quad (6)$$

(7)

Hence the triangle is **not RIGHT** angled.

To Check if the triangle is ISOSCELES

If the triangle is ISOSCELES, a median should be perpendicular to a side of the triangle.

Let the midpoints of the sides **AB**, **BC**, **AC** be **x**, **y** and **z**.

$$\mathbf{x} = \frac{\mathbf{A} + \mathbf{B}}{2} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ \frac{3}{2} \end{pmatrix} \quad (8)$$

$$\mathbf{y} = \frac{\mathbf{B} + \mathbf{C}}{2} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{3}{2} \end{pmatrix} \quad (9)$$

$$\mathbf{z} = \frac{\mathbf{A} + \mathbf{C}}{2} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (10)$$

Theoretical Solution

The vectors of medians of the triangle are:

$$\mathbf{x} - \mathbf{C} = \begin{pmatrix} -2 \\ \frac{3}{2} \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ \frac{3}{2} \end{pmatrix} \quad (11)$$

$$\mathbf{y} - \mathbf{A} = \begin{pmatrix} 2 \\ \frac{3}{2} \end{pmatrix} - \begin{pmatrix} -4 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ \frac{3}{2} \end{pmatrix} \quad (12)$$

$$\mathbf{z} - \mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{2} \end{pmatrix} \quad (13)$$

Theoretical Solution

Then,

$$(\mathbf{x} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -6 & \frac{3}{2} \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -3 \end{pmatrix} = 24 - \frac{9}{2} = \frac{39}{2} \neq 0 \quad (14)$$

$$(\mathbf{y} - \mathbf{A})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} 6 & \frac{3}{2} \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 3 \end{pmatrix} = -24 + \frac{9}{2} = \frac{-39}{2} \neq 0 \quad (15)$$

$$(\mathbf{z} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} 0 & \frac{3}{2} \end{pmatrix} \cdot \begin{pmatrix} -8 \\ 0 \end{pmatrix} = 0 \quad (16)$$

$\therefore$ We can conclude that triangle **ABC** is an **ISOSCELES** triangle.


```
#include <stdio.h>

// Function to check if triangle is right-angled
// Returns 1 if right-angled, 0 otherwise
int is_right_triangle() {
    // Vertices
    float Ax = -4, Ay = 0;
    float Bx = 0, By = 3;
    float Cx = 4, Cy = 0;

    // Side vectors
    float AB[2] = {Ax - Bx, Ay - By};
    float BC[2] = {Bx - Cx, By - Cy};
```

```
float AC[2] = {Ax - Cx, Ay - Cy};

// Dot products
float dot1 = AB[0]*BC[0] + AB[1]*BC[1];
float dot2 = BC[0]*AC[0] + BC[1]*AC[1];
float dot3 = AC[0]*AB[0] + AC[1]*AB[1];

if (dot1 == 0 || dot2 == 0 || dot3 == 0)
    return 1; // Right-angled
else
    return 0; // Not right-angled
}

// Function to check if triangle is isosceles
// Returns 1 if isosceles, 0 otherwise
int is_isosceles() {
```

```
// Vertices
float Ax = -4, Ay = 0;
float Bx = 0, By = 3;
float Cx = 4, Cy = 0;

// Midpoints of sides
float x_mid[2] = {(Ax + Bx) / 2, (Ay + By) / 2};
float y_mid[2] = {(Bx + Cx) / 2, (By + Cy) / 2};
float z_mid[2] = {(Ax + Cx) / 2, (Ay + Cy) / 2};

// Medians
float median1[2] = {x_mid[0] - Cx, x_mid[1] - Cy};
float median2[2] = {y_mid[0] - Ax, y_mid[1] - Ay};
float median3[2] = {z_mid[0] - Bx, z_mid[1] - By};
```

```
// Side vectors
float AB[2] = {Ax - Bx, Ay - By};
float BC[2] = {Bx - Cx, By - Cy};
float AC[2] = {Ax - Cx, Ay - Cy};

// Dot products (median $\perp$ opposite side ?)
float dot1 = median1[0]*AB[0] + median1[1]*AB[1];
float dot2 = median2[0]*BC[0] + median2[1]*BC[1];
float dot3 = median3[0]*AC[0] + median3[1]*AC[1];

if (dot1 == 0 || dot2 == 0 || dot3 == 0)
    return 1; // Isosceles
else
    return 0; // Not isosceles}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load C library
c_lib = ctypes.CDLL('./triangle.so')

# Function signature
triangle_properties = c_lib.triangle_properties
triangle_properties.argtypes = [
    ctypes.POINTER(ctypes.c_double), # sides[3]
    ctypes.POINTER(ctypes.c_double) # angles[3]
```

```
]
triangle_properties.restype = None

# Prepare arrays
sides = (ctypes.c_double * 3)()
angles = (ctypes.c_double * 3)()

# Call the C function
triangle_properties(sides, angles)

# Convert to Python lists
sides_py = [sides[i] for i in range(3)]
angles_py = [angles[i] for i in range(3)]
```

```
print(fSides (AB, BC, CA): {sides_py})
print(fAngles (A, B, C): {angles_py} (degrees))

# --- Plotting ---
A = np.array([-4, 0])
B = np.array([0, 3])
C = np.array([4, 0])

plt.figure(figsize=(8,6))
x_coords = [A[0], B[0], C[0], A[0]]
y_coords = [A[1], B[1], C[1], A[1]]

plt.plot(x_coords, y_coords, 'b-', linewidth=2, label=Triangle
ABC)
```

```
# Mark vertices
plt.scatter([A[0], B[0], C[0]], [A[1], B[1], C[1]], color=red)
plt.text(A[0]-1, A[1]-0.3, fA(-4,0)\nA{angles_py[0]:.2f}, color=
red)
plt.text(B[0]+0.2, B[1]+0.3, fB(0,3)\nB{angles_py[1]:.2f}, color=
green)
plt.text(C[0]+0.4, C[1]-0.3, fC(4,0)\nC{angles_py[2]:.2f}, color=
blue)

# Annotate side lengths at midpoints
midpoints = [
    ((A[0]+B[0])/2, (A[1]+B[1])/2, sides_py[0]),
    ((B[0]+C[0])/2, (B[1]+C[1])/2, sides_py[1]),
    ((C[0]+A[0])/2, (C[1]+A[1])/2, sides_py[2])
]
```



```
for (mx, my, d) in midpoints:
    plt.text(mx+0.2, my+0.2, f'{d:.2f}', fontsize=10, color=purple)

# Axis
plt.axhline(0, color=black, linewidth=0.5)
plt.axvline(0, color=black, linewidth=0.5)
plt.axis('equal')
plt.grid(True)
plt.title(Triangle ABC with Sides and Angles (from C))
plt.legend()
plt.savefig(fig_3.png)
plt.show()
```

